Vijay Academy Presents

Savitribai Phule Pune University

310251 – Data Science and Big Data Analytics (DSBDA)

TE Computer Engineering

What You Get : Strategy For Exam :

- Simple Explanation
- PPT + Notes
- PYQs + Most IMP Questions

Unit III : Big Data Analytics Life Cycle

1) MOST IMPORTANT (Repeated 6–7 times)

हे topics Q1/Q2 मध्ये दर वर्षी nearly compulsory येतात.

1. Data Analytics Life Cycle (Full diagram + 6 phases) [7]

Discovery, Data Preparation, Model Planning, Model Building, Communicate Results, Operationalize

2. Discovery Phase – Activities / Identifying Data Resources [6]

business problem, data sources, constraints, analytic approach

3. Data Preparation Phase (ETLT + Analytics Sandbox) [6]

ETLT: Extract → Transform → Load → Transfer, sandbox, data cleaning steps

4. Model Planning Phase – Activities + Tools [5]

selecting analytic technique, choosing model type, feature selection

tools: R, Python, SAS, WEKA, RapidMiner

5. Model Building Phase [5]

training/testing split, iterative model development, evaluation

6. Sources of Big Data (Structured / Unstructured / Semi-Structured) [5]

7. Big Data Analytics Architecture (diagram + components) [5]

2) MEDIUM IMPORTANT (Asked 3–4 times)

1. Characteristics of Big Data (3 V's or 5 V's) [3]
Volume, Velocity, Variety
(sometimes Veracity, Value)

2. Applications of Big Data Analytics [3]

3. Model Selection for Data Analytics [3]
Criteria, metric-based comparison

4. BI vs Data Science [3]

5. Big Data Driving Factors / Data Deluge [2-3]

6. Tools for Model Building (short note) [2-3]

3) LOW / NOT ASKED (0–1 time OR rarely)

1. Communication of Results Phase [1]

2. Operationalize Phase

3. Business Intelligence Architecture

4. Detailed diagrams of each phase

Unit 4 (Predictive Big Data Analytics with Python)

1) MOST IMPORTANT (Repeated 6–7 times)

1. Apriori Algorithm (with example) [6]
Frequent itemsets, Support & Confidence, Step-by-step working

2. Naïve Bayes Classifier (theory + example) [6]
Formula, Assumptions, Applications, Small dataset calculation (spam/loan examples)

3. Linear Regression (definition + working + objectives) [6]
simple vs multiple, cost function, evaluation metrics (MSE, R^2)

4. Logistic Regression [6]
sigmoid function, need/use, difference vs linear regression

5. Data Preprocessing Steps [5]
removing duplicates, handling missing values,transformation / mapping
replacing values

6. Information Gain + Entropy [4]
formula, example calculation

7. Decision Tree (construction using entropy/IG) [5]
splitting criteria
example tree

2) MEDIUM IMPORTANT (Asked 3–4 times)

1. Types of Analytics (Descriptive, Predictive, Prescriptive) [3-4]

2. Python Essential Libraries [3]

NumPy, Pandas, Matplotlib, Scikit-learn

3. Classification Algorithms Overview [3]

Naive Bayes, Decision Tree

4. scikit-learn basics [2-3]

installing
dataset loading
missing value filling
regression/classification code

5. FP-Growth Algorithm [2]

(Rarely but sometimes asked)

3) NOT ASKED / LOW PRIORITY (0–1 time only)

1. Detailed Python code for preprocessing

Only theory asked.

2. Matplotlib advanced plots

Line plot, scatter, etc. rarely in Unit 4.

3. Deep theory of mapping/encoding

Only short explanations asked.

4. Full FP-Growth derivation

Very rare.

5. Logistic regression matrix form

Not asked.

UNIT 5 — Big Data Analytics & Model Evaluation

1) MOST IMPORTANT (REPEATED 6–7 TIMES)

1. K-Means Clustering Algorithm (with numerical example) [7]
   2 clusters example, 3 clusters example, Find new centroids, Show iteration-1 result

2. Text Preprocessing Steps [6]
   Tokenization, Stopword removal, Stemming, Lemmatization, POS tagging

3. TF–IDF (theory + example) [6]
   Term Frequency, Inverse Document Frequency, Simple example

4. Confusion Matrix + Metrics [6]
   Metrics asked : Accuracy, Precision, Recall, Error rate, F1-score

5. Holdout Method / Train–Validation–Test Split [5]

2) MEDIUM IMPORTANT (3–4 TIMES ASKED)

1. Hierarchical Clustering (Agglomerative & Divisive) [4]

2. Time Series Analysis [3-4]
Trend, Seasonality, Components

3. Bag of Words (BoW) [3]

4. AUC–ROC Curve [3]
ROC, True Positive Rate, False Positive Rate

5. k-Fold Cross Validation [3]

6. Social Network Analysis (definition + applications) [3]

3) NOT ASKED / RARELY ASKED (0–1 TIME)

1. Business analytics introduction

(Almost never asked directly)

2. Detailed time-series forecasting formulas (ARIMA etc.)

Only basic explanation asked, never deep math.

3. sklearn code for clustering/time-series

Not asked in theory papers.

4. Elbow plot (rare)

(Only 1 paper out of 7)

Unit 6 (Data Visualization & Hadoop)

1) MOST IMPORTANT (Repeated 6–7 times)

1. Hadoop Ecosystem (diagram + components) [7]
HDFS, MapReduce, YARN, Hive, Pig, HBase

2. MapReduce (definition + working) [6]
Map Phase, Reduce Phase, Example

3. Pig (short note / architecture) [6]

4. Hive (architecture + features / short note) [6]

5. Data Visualization – Definition + Objectives [6]

6. Challenges of Big Data Visualization [6]

2) MEDIUM IMPORTANT (Asked 3–4 times)

1. Types of Data Visualization [4]
Bar, Line, Scatter, Pie, Map

2. Data Visualization Tools [4]
Tableau, Power BI, Matplotlib

3. Applications of Data Visualization [3]

4. Python Plots [3]
Line plot, Scatter plot, Histogram, Box plot

5. Tableau (short note / features / uses) [3]

6. Power BI or Qlik (short note) [2]

3) NOT ASKED / LOW PRIORITY (0–1 time)

1. Detailed Big Data visualization techniques (advanced)

Not asked in depth.

2. Density plot (rare)

Only 1 paper.

3. Analytical techniques used in big data visualization

Rare.

4. Detailed working of HBase, Sqoop, Flume

Not in theory questions.

5. Hive query examples

Not asked.

Thank You!

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